Skills Summary

Composing Functions from Symbolic Rules

Use this reference while completing your worksheet or when studying.

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1. Run a composition at a number, inside rule first

Notation. $(f \circ g)(x)$ means $f\big(g(x)\big)$. The rule written next to the input — here g — is the *inner* rule, and it always runs first.

At a number: compute $g(\text{number})$, then feed that single output into f. Two machines in a row, never both at once.

Example: $f(x) = x - 6$ and $g(x) = 5x$. Find $(f \circ g)(4)$.
Step 1. The circle notation puts g next to the input, so g runs first and its output becomes the input of f.
Step 2. $g(4) = 5 \cdot 4 = 20$, and then $f(20) = 20 - 6$.
$(f \circ g)(4) = \mathbf{14}$

Try it: $f(x) = 2x$ and $g(x) = x + 3$. Find $(f \circ g)(5)$.

check: **16**

2. Build the composed rule from two symbolic rules

The substitution rule. Wherever the outer rule shows its input variable, write the *whole* inner rule in parentheses — in every slot, not just the first one. Only then expand and combine like terms.

$f(x) = x^2 + 2x$ with input $\square$ becomes $f(\square) = \square^2 + 2\square$.

Example: $f(x) = x^2 + 2x$, $g(x) = x - 1$. Write $(f \circ g)(x)$.
Step 1. g is inner, so every input slot of f takes $(x - 1)$ — and f has two slots, one squared and one linear.
Step 2. $(x-1)^2 + 2(x-1) = x^2 - 2x + 1 + 2x - 2$, and the $-2x$ and $+2x$ cancel.
$(f \circ g)(x) = \boldsymbol{x^2 - 1}$

Try it: $f(x) = x^2 - 4$, $g(x) = x + 2$. Write $(f \circ g)(x)$ in simplest form.

check: $\boldsymbol{x^2 + 4x}$

3. Order matters: $f(g(x))$ is not $g(f(x))$

Swapping which rule is inner usually changes the function, so $f \circ g$ and $g \circ f$ must be built separately.

Quick test: pick one input and run it both ways. If the two outputs differ, the two compositions are different rules — one counterexample settles it.

Example: $f(x) = 3x - 2$, $g(x) = x^2 + 1$. Build both orders.
Step 1. Each order names a different inner rule, so do two separate substitutions rather than reusing the first answer.
Step 2. $f(g(x)) = 3(x^2+1)-2 = 3x^2+3-2$, while $g(f(x)) = (3x-2)^2+1 = 9x^2-12x+4+1$.

$$(f \circ g)(x) = \mathbf{3x^2 + 1} \quad \text{but} \quad (g \circ f)(x) = \mathbf{9x^2 - 12x + 5}$$

Try it: $f(x) = x + 5$, $g(x) = x^2$. Write $(g \circ f)(x)$ expanded.

check: $\boldsymbol{x^2 + 10x + 25}$

4. Inverses: composition hands the input straight back

Two rules are inverses exactly when composing them — *both ways* — returns the input untouched: $f\big(g(x)\big) = x$ and $g\big(f(x)\big) = x$. One direction alone is not the definition, so always check the pair.

Example: Are $f(x) = 2x - 6$ and $g(x) = \frac{x+6}{2}$ inverses?
Step 1. The definition is a composition statement, so substitute rather than compare the two formulas by eye.
Step 2. $f(g(x)) = 2\left(\frac{x+6}{2}\right) - 6 = (x+6) - 6$, because the factor 2 cancels the division.

$$f\big(g(x)\big) = \boldsymbol{x}, \text{ and } g\big(f(x)\big) \text{ works out the same way}$$

Try it: $f(x) = 5x + 1$, $g(x) = \frac{x-1}{5}$. Simplify $g\big(f(x)\big)$.

check: $\boldsymbol{x}$

(!) Watch out: composition is not multiplication. $(f \circ g)(x)$ is $f\big(g(x)\big)$, while $f(x) \cdot g(x)$ multiplies the two outputs — different operations, different answers. And when you substitute a whole rule, keep it in parentheses: forgetting them turns $(x-3)^2$ into $x - 3^2$.